

ALPHA DATA

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Higher Colleges of Technology

RFP Ref. NE-26-0129 — Secure Access Service Edge (SASE) — 3 Years

SASE Architecture & Design

Fortinet FortiSASE — Unified Cloud SASE for 29,000+ Users Across 17 Campuses

Portal Section 3 — Item 1 (Evaluation Criterion 3.1, Weight 15%)

Technical Submission — September 2026

Commercial-in-Confidence

Alpha Data proposes Fortinet FortiSASE: a single-vendor, fully cloud-delivered SASE that converges SWG, FWaaS, RBI/Enterprise Browser, DNS Security, inline-CASB and Universal ZTNA on one operating system (FortiOS), one unified agent (FortiClient) and one management console — sized for HCT's 29,022 users plus Masfout and 15% year-on-year growth, delivered through in-region PoPs meeting HCT's 99.999% availability and ≤25 ms latency SLAs, and migrated from the incumbent Zscaler ZIA and on-premises proxy platforms within the mandated 3-month window.

1. Understanding of HCT's Requirement

HCT operates the UAE's largest applied higher-education institution: 26,303 students and 2,719 staff across 16 campuses plus the new Masfout campus (adding ~100 students and 50 staff), a Primary Data Centre in Abu Dhabi (8 Gbps breakout) and a DR Data Centre in Dubai, with ~39.5 Gbps of aggregate local internet breakout. Today, internet security is split across two legacy platforms — Zscaler ZIA for managed users and an on-premises proxy for BYOD/student users — whose licences and support expire on 31 December 2026. HCT requires a single cloud SASE platform that secures all internet-bound traffic for both managed and BYOD populations, integrates with the existing zero-trust ecosystem (IdP, NAC, NGFW, EDR/XDR, SIEM/SOAR, UEM), and is fully deployed, migrated and commissioned within 3 months of award — with phasing as the delivery method, not an extension of the deadline.

Three realities shape our design: (1) students access services on-campus only, while staff/faculty require a managed remote flow; (2) cutovers are permitted on weekends only, outside examination periods; and (3) the platform must absorb 15% year-on-year growth in users and devices within the fixed 3-year price. The architecture below is engineered around these constraints.

2. Proposed Solution at a Glance

SOW Requirement	How FortiSASE Delivers
Fully cloud-delivered SASE (3.1.1)	FortiSASE: cloud-native security fabric on FortiOS, delivered from geo-redundant PoPs; no reliance on existing HCT infrastructure; all components newly procured, latest generation, no EOL/EOS
Converged SWG + FWaaS + RBI + DNS Security (3.1.2)	All four pillars native to one platform, one policy engine, one console — plus inline-CASB and DEM
Agent + agentless models (3.1.4, 3.6, 3.7)	FortiClient unified agent for managed devices (Windows 10/11, macOS, iOS, Android); agentless BYOD via explicit proxy/PAC, DNS steering, captive portal and certificate-based authentication
AI-powered security (3.5)	FortiGuard AI: inline ML anti-malware, AI phishing detection, UEBA risk scoring, encrypted-traffic analysis, GenAI application discovery and granular per-app controls, AIOps and predictive capacity
Campus breakout steering (3.1.10)	IPsec tunnels from campus edge to nearest FortiSASE PoP; FortiSASE Branch On-Ramp licensed for 20 locations covering all 17 campuses + Primary DC + DR
Ecosystem integration (3.9)	SAML 2.0/OIDC/SCIM to Entra ID, RADIUS accounting for NAC, IPsec + log sharing with NGFW, EDR/XDR posture, SIEM via syslog/CEF, UEM/Intune posture, STIX/TAXII threat intel, ITSM API
Optional ZTNA — same platform (3.25)	Universal ZTNA native to FortiSASE and the same FortiClient agent — same vendor, same console, same policy engine as mandated by HCT

SOW Requirement	How FortiSASE Delivers
	clarification Q.5; 200 concurrent sessions
Sandbox tenant (3.16.10)	Dedicated non-production tenant mirroring production, full feature/policy parity, provided for the full 3-year term at no additional cost

3. Bill of Materials (Technical) — Two Options Offered

Alpha Data offers HCT two technically validated options. Both license every HCT user on FortiSASE Advanced (full SWG, FWaaS, IPS, DNS Security, SSL inspection, anti-malware, CASB, DEM and unified-agent ZTNA/EPP, 3 devices per user, FortiCare Premium) and both include the FortiGate-401F secure edge at the data centres. They differ only in how campus/site traffic reaches the FortiSASE cloud. Both options are priced separately in the commercial submission.

Option 1 — FortiSASE with Branch On-Ramp (BOR) — Recommended

SKU	Description	Qty	Notes
FC5-10-EMS05-676-02-DD	FortiSASE Advanced Subscription, 10,000+ users — Fortinet cloud-delivered security (IPS, Web, DNS, SSL, Anti-Malware, CASB) plus enterprise-ready NOC/SOC integrations, secure private access via unified agent (SASE, ZTNA, EPP) or agentless via SWG; 3 devices per user; FortiCare Premium	29,022	All HCT users; Masfout adds a further 50 staff managed users, absorbed within growth headroom
FC1-10-EMS05-769-02-36	FortiSASE Branch On-Ramp, 3-year — per location, 1 Gbps of shared bandwidth for up to 2,000 FortiGate or 3rd-party IPsec connections to Fortinet Cloud locations (2-20 locations)	20	20 per account (limit); aggregate bandwidth across campuses ~39.5 Gbps; each on-ramp limited to 1 Gbps aggregate
FC-10-0401F-1389-02-36	FortiGate-401F, 3-Year SD-WAN Service Bundle — SD-WAN underlay & application monitoring, FortiGuard SD-WAN SLA database updates, overlay orchestration, FortiSASE 10-user starter kit, SASE Secure Private Access (SPA) Connector, FortiTelemetry Cloud, 1-year cloud log retention	3	2 x Primary DC (HA) + 1 x DR — new hardware, latest generation, no EOL/EOS

Option 2 — FortiSASE Only

SKU	Description	Qty	Notes
FC5-10-EMS05-676-02-DD	FortiSASE Advanced Subscription, 10,000+ users — identical scope to Option 1	29,022	All HCT users
FC-10-0401F-1389-02-36	FortiGate-401F, 3-Year SD-WAN Service Bundle — identical scope to Option 1	3	2 x Primary DC (HA) + 1 x DR

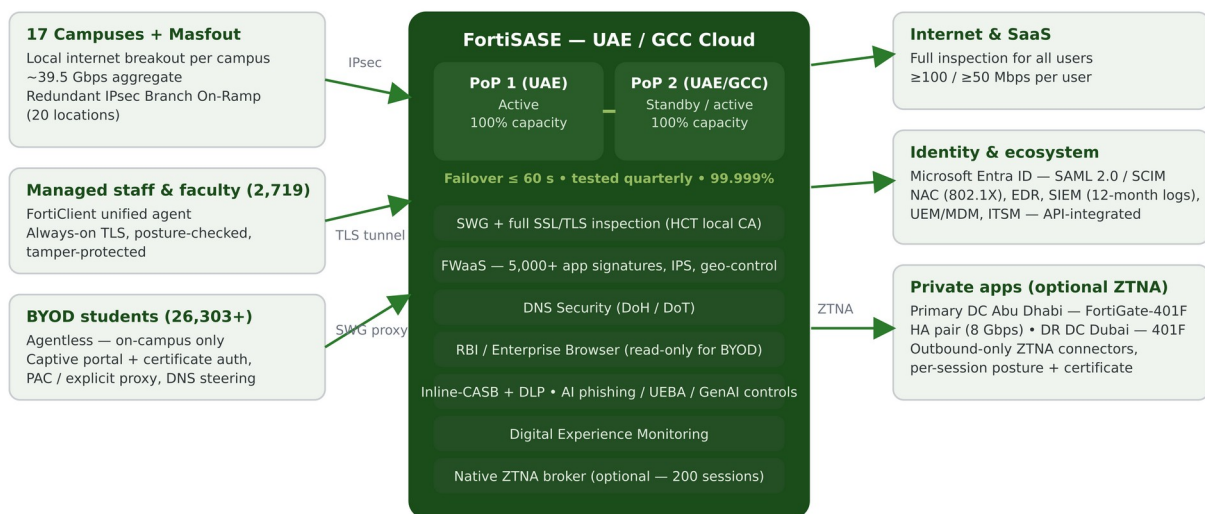
How the two options compare

- Option 1 (recommended): dedicated 1 Gbps IPsec Branch On-Ramps at all 20 locations give deterministic, per-campus tunnel capacity aligned to the Appendix-A link sizes and the ~39.5 Gbps aggregate breakout — every device on campus (including unmanaged IoT, labs and guest segments) is steered and inspected at site level without an agent, and each on-ramp carries its own 50,000 GB/year data-transfer entitlement in addition to the 250 GB/year per user.
- Option 2 (cost-optimised): campus traffic reaches FortiSASE through per-user steering only — the FortiClient agent for managed users and PAC/explicit-proxy/DNS steering for BYOD — with site-to-cloud IPsec terminating on the FortiGate-401F SPA connectors at the data centres where private access is needed. Lower subscription cost, with cloud on-ramp capacity drawn from the per-user service rather than dedicated site tunnels.
- Under both options every SOW security, SLA, migration, managed-service and governance commitment in this submission applies unchanged; the choice affects only the campus transport model and commercial total. Alpha Data recommends Option 1 for HCT's scale of agentless BYOD and its deterministic per-site throughput; Option 2 remains fully supported should HCT prioritise cost.

4. Architecture Design

Figure 1 — HCT SASE target architecture: all user communities and all 17 campuses converge on dual UAE/GCC FortiSASE PoPs

HCT SASE Target Architecture — Fortinet FortiSASE (single platform, single agent, single console)



All components new and latest generation — no EOL/EOS, no dependency on HCT's existing Zscaler or on-premises proxy.

4.1 Traffic Flows

- Managed users (staff/faculty, on-campus and remote): FortiClient unified agent steers traffic over always-on TLS/IPsec to the nearest UAE-region FortiSASE PoP for full inspection (SWG, FWaaS, IPS, AV/sandbox, DLP, CASB). Fail-open/fail-close behaviour is configurable per Section 3.7.2; tamper protection prevents user bypass.
- BYOD users (students, on-campus only): agentless enrolment via captive portal, PAC/explicit proxy and DNS-layer steering with certificate-based authentication; browser sessions steered to the SASE PoP; read-only RBI available for sensitive web applications (3.4.3).

- Campus/site traffic: each campus' local internet breakout is steered via redundant IPsec tunnels (Branch On-Ramp) to the nearest PoP, sized per Appendix-A link capacities, with automatic failover between tunnels and PoPs.
- Data centres: FortiGate-401F pairs at the Primary DC (HA) and DR provide the secure edge/SD-WAN underlay and serve as ZTNA application gateways for the optional private-access scope.

Figure 2 — Managed staff/faculty traffic flow (FortiClient agent, on-campus and remote)

Traffic Flow — Managed Staff / Faculty (FortiClient agent, on-campus and remote)

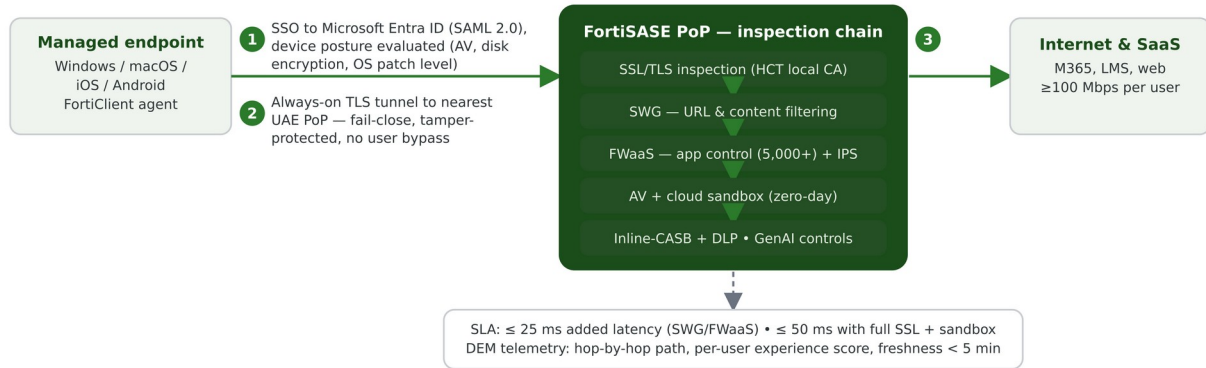


Figure 3 — BYOD student traffic flow (agentless SWG, on-campus only)

Traffic Flow — BYOD Students (agentless, on-campus only)

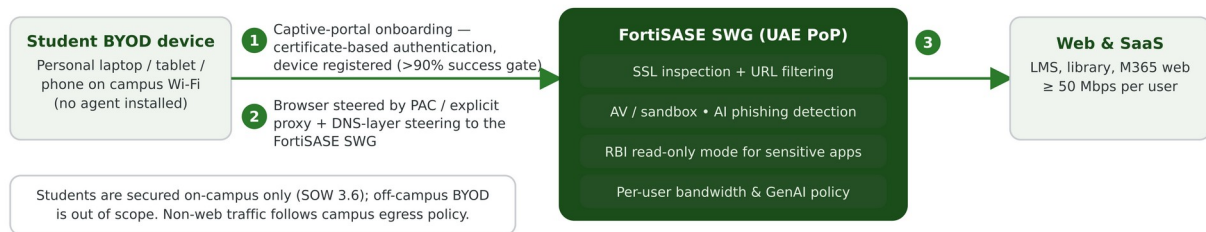
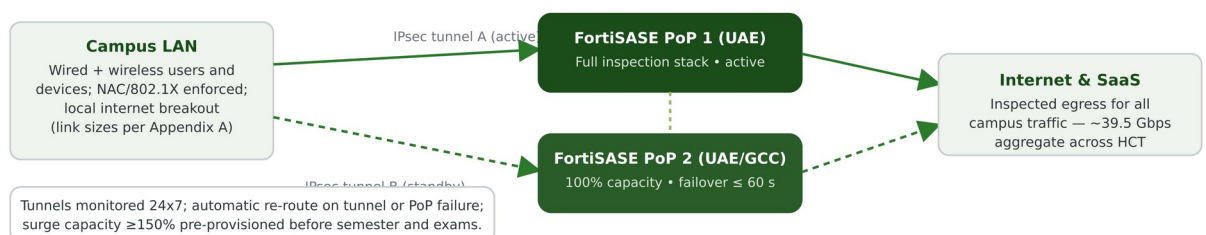


Figure 4 — Campus site traffic flow via redundant IPsec Branch On-Ramp with dual-PoP failover

Traffic Flow — Campus Sites via IPsec Branch On-Ramp (20 locations)



4.2 Security Services (Sections 3.2–3.5)

- SWG: inline proxy with full SSL/TLS 1.2/1.3 inspection using HCT's local CA-signed intermediate certificate; 80+ URL categories over millions of rated domains; multi-engine AV, heuristics and cloud sandboxing; file-type and MIME controls; per-application bandwidth management; proxy-avoidance prevention.
- FWaaS: FortiOS NGFW-equivalent controls in the cloud — L7 application identification across 5,000+ application signatures with sub-application granularity (e.g. allow Teams meetings, block Teams file transfer), IPS, geo-blocking and logging of all sessions.

- RBI / Enterprise Browser: policy-driven isolation by URL category, user risk score or device posture; pixel-streaming with no active content reaching the endpoint; read-only mode for BYOD; clientless browser-based access for institutional web apps.
- DNS Security: inspection of DNS, DoH and DoT; blocking of DNS tunnelling, DGA, newly registered and known-malicious domains.
- AI-powered protection: FortiGuard inline ML anti-malware and AI phishing detection; UEBA with dynamic per-user/per-device risk scores feeding policy; encrypted traffic analysis (JA3/JA4, certificate attributes) without decryption where decryption is not possible; GenAI discovery with per-application, per-audience controls (e.g. ChatGPT allowed for staff, read-only for students; uploads to unsanctioned AI blocked) and AI usage analytics dashboards; AIOps root-cause analysis and predictive capacity forecasting for academic peaks.

4.3 Scalability, Capacity and Performance (Section 3.8)

- Licensed for 29,022 users at award; the subscription model scales linearly to absorb the indicative 15% year-on-year user/device growth within the fixed 3-year price, per HCT clarification Q.1/Q.3.
- 45,000 concurrent devices (3 devices per user) supported; elastic cloud PoP capacity pre-provisioned to $\geq 150\%$ of normal peak at least 7 days before semester starts, examinations, and graduation events (3.11.29.2).
- Performance engineered to HCT SLAs: ≤ 25 ms added latency for SWG/FWaaS, ≤ 50 ms for full SSL inspection plus sandboxing; ≥ 100 Mbps per managed user and ≥ 50 Mbps per BYOD user through the full inspection stack; 99.999% platform availability with financially backed credits.
- Regional resilience: a minimum of two geographically separated UAE/GCC PoPs, each independently capable of 100% of HCT traffic, with automatic failover within 60 seconds and quarterly failover testing (3.20.2).
- Digital Experience Monitoring: hop-by-hop path visibility (Wi-Fi $\rightarrow$ LAN $\rightarrow$ WAN $\rightarrow$ PoP $\rightarrow$ application), synthetic probes to HCT's top SaaS destinations, per-user/per-application experience scoring and AI-driven root-cause analysis; DEM data freshness under 5 minutes.

4.4 Integration Architecture (Section 3.9)

HCT System	Integration Method
Identity (Microsoft Entra ID / AD)	SAML 2.0 / OIDC SSO with MFA; SCIM provisioning; AD/Exchange/Terminal Server agents; RADIUS accounting for wireless; captive portal for wired
NAC	RADIUS accounting integration; posture exchange for policy decisions
NGFW (campus edge)	IPsec tunnel steering to PoPs; bidirectional log sharing
EDR/XDR	Endpoint posture (agent status, patch level, encryption) consumed in access policy
SIEM / SOAR	Real-time log forwarding via syslog (RFC 5424) / CEF; RESTful APIs with OpenAPI documentation
UEM / MDM (Intune, Jamf, WS1)	Device compliance queried as policy input; silent agent deployment and update
Threat Intelligence	STIX/TAXII 2.1 feed ingestion; institutional IoC blocklists
ITSM (ServiceNow / IVANTI)	API-driven automated ticket creation for security alerts and policy violations

HCT System	Integration Method
Microsoft DLP & Defender for Cloud Apps	Consistent data-protection policy enforcement and SaaS risk-based access via native Microsoft integrations
HSM / key management (3.16.4.6)	Supports FIPS 140-2 Level 2+ validated key stores for inspection/SAML/API keys; dynamic in-cloud certificate signing for inline SSL inspection, with HCT-controlled intermediate CA

4.5 Management, Reporting and Non-Production (Sections 3.10, 3.16.10)

- Single HTML5 cloud console (no Java/Flash/ActiveX) with RBAC, NOC/SOC dashboards, IOC scanning and prioritised compromised-host lists.
- 12 months of searchable log retention with archival options and continuous forwarding to HCT's SIEM independent of online retention.
- Comprehensive REST APIs (policy, users, logs, configuration) plus JSON/XML web services for third-party integration; quarterly configuration/policy export in vendor-neutral JSON/XML for HCT's data escrow (3.23.6).
- Dedicated sandbox/staging tenant with full feature and policy parity — isolated from production so no test impacts live users — provided from deployment through the full contract term at no additional cost, used for policy validation, upgrade testing and HCT administrator training.

5. Migration and Deployment Plan (Section 3.15)

Full population — every user, every campus — migrated, validated and commissioned within 3 months, using parallel weekend migration groups, with the legacy platforms kept fully operational through parallel-run and for 60 days after HCT's written cutover confirmation.

Phase 0 — Mobilisation & Environment Assessment (Weeks 1-2)

- Bidder-led assessment of both legacy platforms (Zscaler ZIA for managed users; on-premises proxy for BYOD), covering policy scale, custom URL categories, decryption rules, endpoint-agent coexistence and export formats — Alpha Data owns migration scoping per HCT clarification Q.15, with remediation of any post-award discoveries at our cost.
- Pre-deployment performance baselining at every campus (latency to top-20 destinations, throughput, top-10 SaaS response times, DNS resolution, DEM scores) per 3.15.8.

Phase 1 — Foundation Build & PoC (Weeks 2-5)

- Production + sandbox tenants provisioned; IdP/NAC/NGFW/SIEM/UEM integrations; HCT intermediate CA onboarding for SSL inspection; policy translation and consolidation from both legacy platforms into a single unified policy set.
- Proof of Concept executed against every Section 3.24.2 acceptance criterion (see submission S3.6), including a rollback drill on a non-production segment before any production migration (3.15.7.4).

Phase 2 — Parallel Site-by-Site Migration (Weeks 5-12)

- Parallel migration groups (multiple campuses per weekend) sequenced with HCT change approval; cutovers on weekends only, never during examination periods, honouring the change-freeze windows of 3.11.29.3.

- Per-site runbook: IPsec on-ramp establishment → managed-agent silent deployment via UEM (>95% success target) → BYOD captive-portal onboarding (>90% target) → validation against baseline → HCT sign-off.
- Rollback triggers per 3.15.7.2 continuously monitored; legacy platforms remain fully operational throughout parallel-run and for a minimum of 60 days after HCT's formal written confirmation (3.15.7.3).

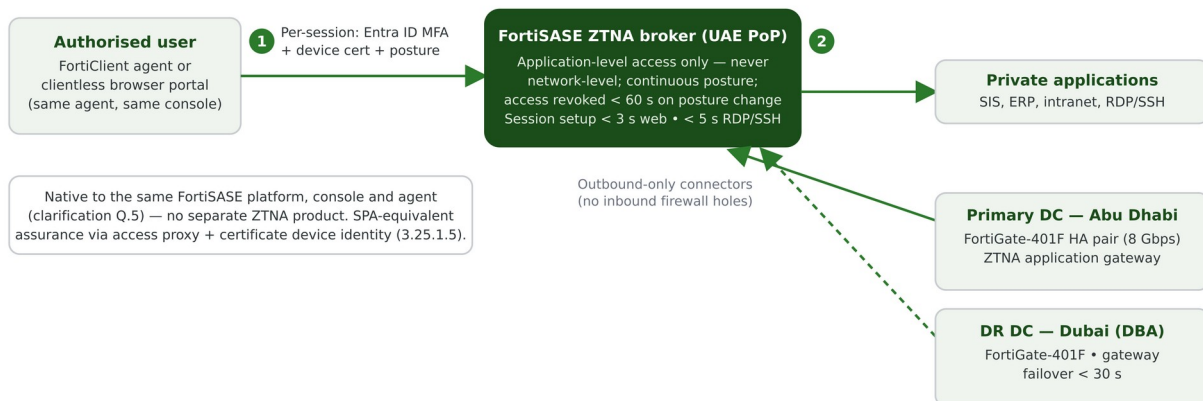
Phase 3 — Commissioning, Handover & Decommissioning (Weeks 12–13+)

- Formal project-to-operations handover with all Section 3.17.6 sign-offs: integration validation, monitoring, configuration baseline, runbooks, escalation contacts, backup/restore drill, SLA baseline and zero open P1/P2.
- 90-day knowledge transfer with practical assessment; OEM training for minimum 4 HCT IT staff; legacy decommissioning only upon HCT's explicit written authorisation.

6. Optional Scope — Zero Trust Network Access (Section 3.25)

Figure 5 — Optional native ZTNA private-access flow to the Primary (Abu Dhabi) and DR (Dubai) data centres

Traffic Flow — Optional Native ZTNA Private Access (200 concurrent sessions)



- ZTNA is native to the same FortiSASE platform, agent and console — no separate product, licence portal or agent, exactly as HCT mandated in clarification Q.5.
- 200 concurrent sessions replacing legacy client VPN, with per-application least-privilege access; deny-all by default; applications individually published — no network-level access.
- HA ZTNA application gateways in the Primary DC (Abu Dhabi) and DR DC (Dubai) on the proposed FortiGate-401F units, hosted on HCT's Hyper-V/Azure environment where connector form factors are preferred; all connectivity outbound-initiated — no inbound firewall rules or DMZ exposure.
- mTLS end-to-end; continuous posture re-evaluation with mid-session termination on posture failure (<60 s); web, thick-client and RDP/SSH application types; clientless browser portal for designated web apps; session recording and PAM integration for privileged access; JIT time-limited grants.
- Two clauses are transparently declared as partial in the compliance matrix (3.25.1.2 agent-connector topology and 3.25.1.5 SPA mechanism): FortiSASE achieves the same security outcomes — application invisibility to unauthorized users and brokered, inspected access — through its ZTNA access-proxy architecture with certificate-based device identity rather than UDP port-knocking SPA. Details and compensating controls are documented in the compliance sheet.

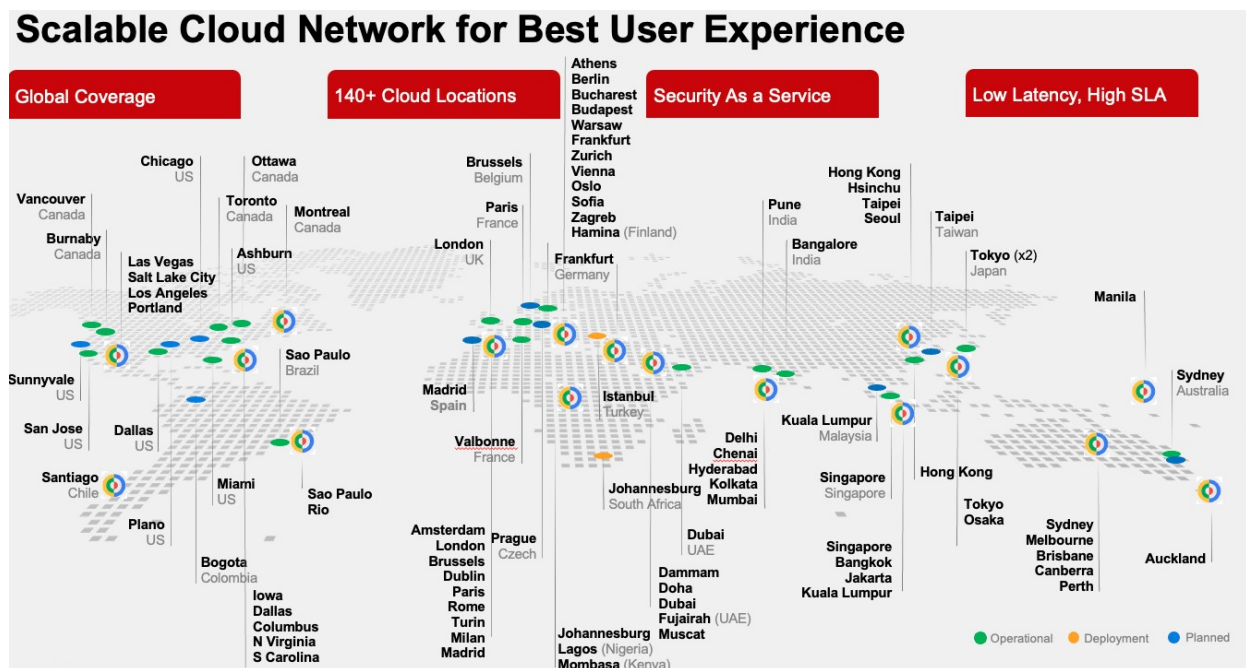
Annex B — OEM Reference Write-up: FortiSASE Architecture & Design Examples

The following architecture reference and design examples are reproduced from the Fortinet FortiSASE technical write-up accompanying this proposal, with the original figures included as published. They evidence that every traffic-flow pattern proposed for HCT in Section 4 — agent-based SIA, agentless SWG, site-based on-ramps, ZTNA and secure SaaS access — is a standard, documented FortiSASE design, not a custom construct.

B.1 FortiSASE Security PoPs

FortiSASE Security PoPs are distributed across the globe; the locations shown with full Fortinet setup are complemented by additional security PoPs delivered under Fortinet's partnership with Google Cloud. The current list of active PoPs is published at <https://docs.fortinet.com/document/fortisase/latest/reference-guide/663044/global-data-centers>. For HCT, the tenant is provisioned exclusively on the two HCT-approved UAE/GCC PoPs.

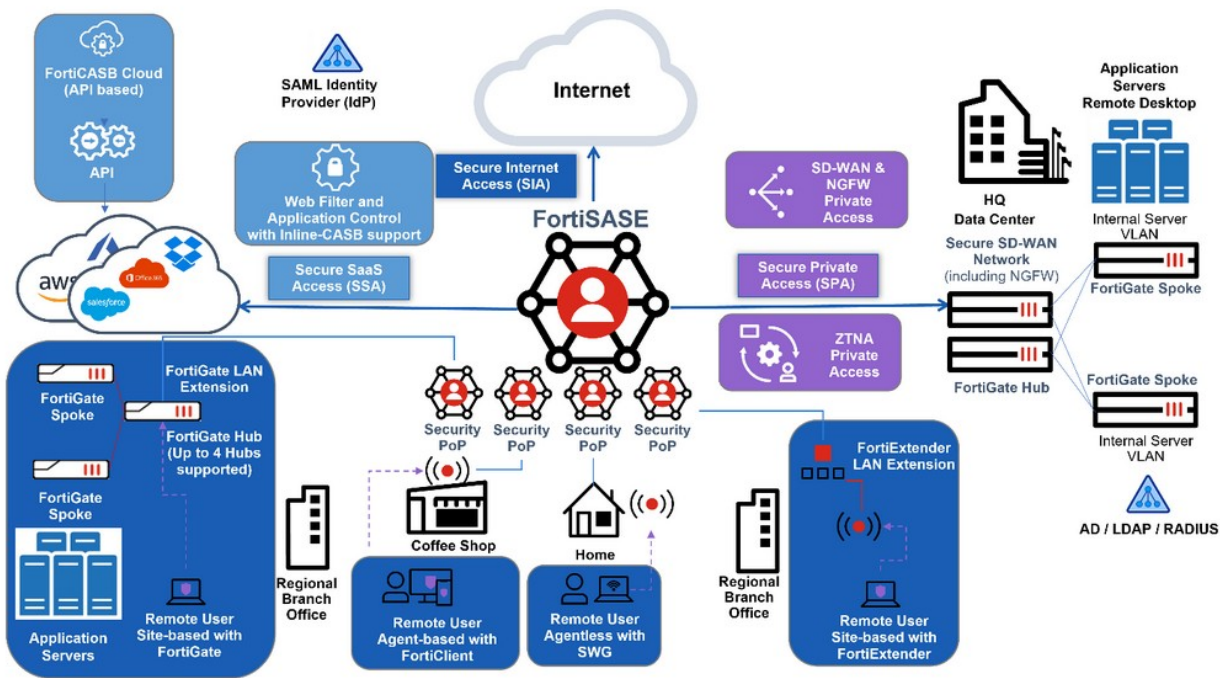
Figure B-1 — FortiSASE global Security PoP coverage (Fortinet original)



B.2 FortiSASE Common Use Cases

FortiSASE addresses secure internet access (SIA), secure private access (SPA) and secure SaaS access (SSA) for agent-based and agentless users from a single cloud platform — the use-case set applied to HCT's staff, students and campuses in Section 4.

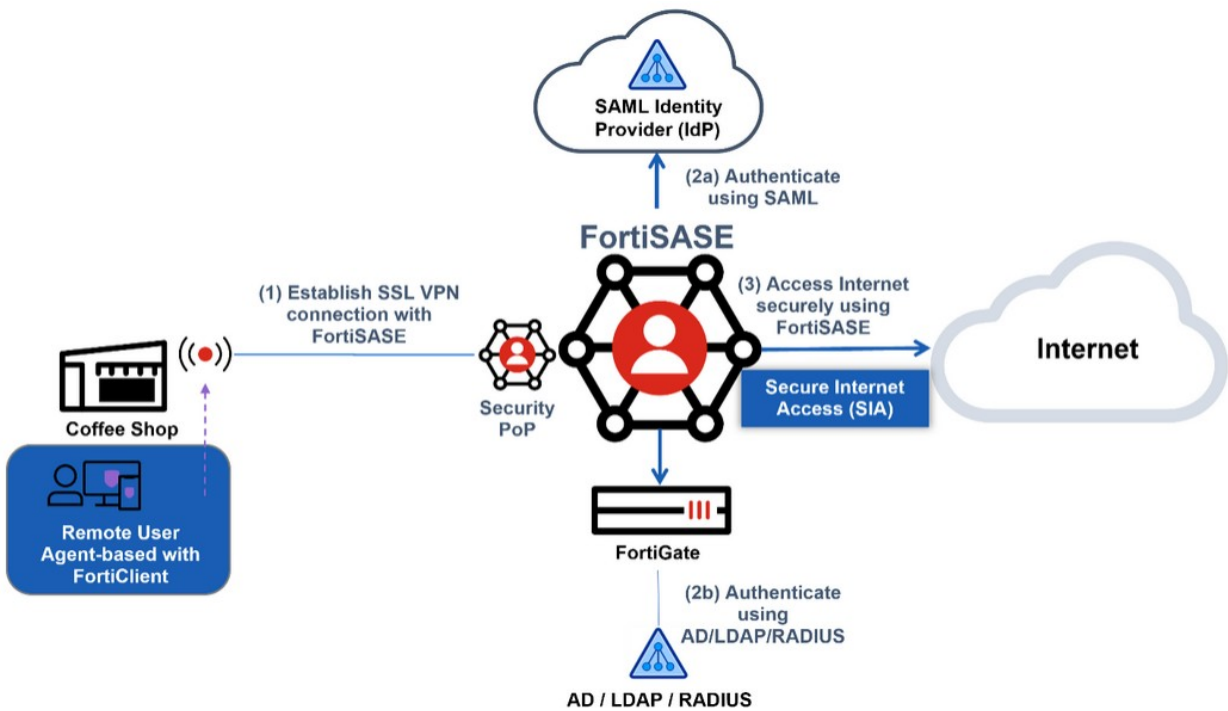
Figure B-2 — FortiSASE common use cases (Fortinet original)



B.3 SIA for agent-based remote users → HCT managed staff/faculty

Secure Internet Access (SIA) for agent-based users involves installing and configuring FortiClient on supported endpoints (Windows, macOS, Linux). The FortiSASE firewall-as-a-service sits between the endpoint and the Internet: FortiClient establishes a full-tunnel VPN to the FWaaS, so agent-based SIA secures all Internet traffic and protocols using VPN policies, with each endpoint connecting to a Security PoP. Authentication is via Active Directory/LDAP, RADIUS or a SAML identity provider, and initial endpoint configuration can be automated with an MDM tool. This is the design applied to HCT’s 2,719 managed staff and faculty (Figure 2).

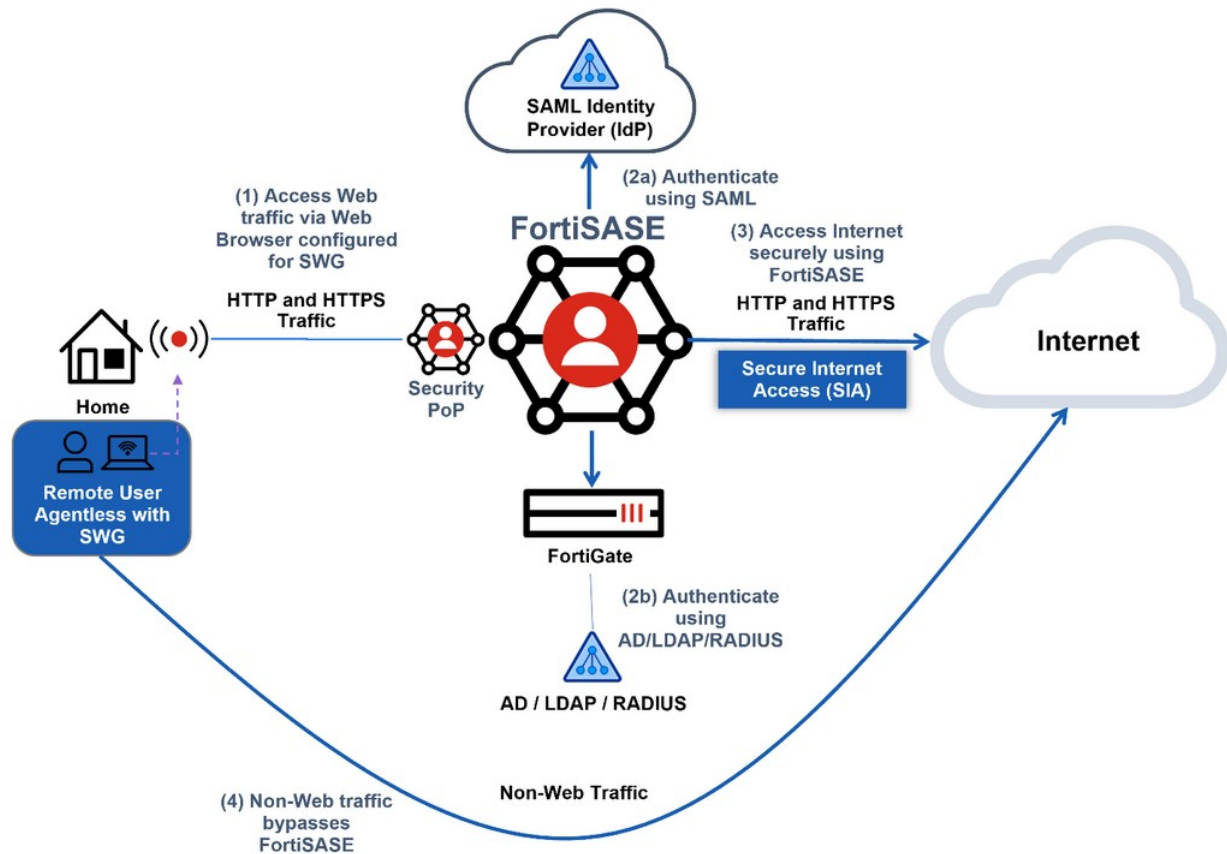
Figure B-3 — SIA for agent-based remote users, typical topology (Fortinet original)



B.4 SIA for agentless remote users → HCT BYOD students

SIA for agentless users configures the web browser — or a browser-based device using a proxy auto-configuration (PAC) file — to use the FortiSASE SWG as an explicit web proxy. HTTP/HTTPS traffic is redirected to the SWG and secured by SWG security policies. Authentication is via AD/LDAP, RADIUS or SAML, and browser proxy settings can be automated (e.g. via GPO). This is the design applied to HCT's 26,303+ BYOD students on campus (Figure 3).

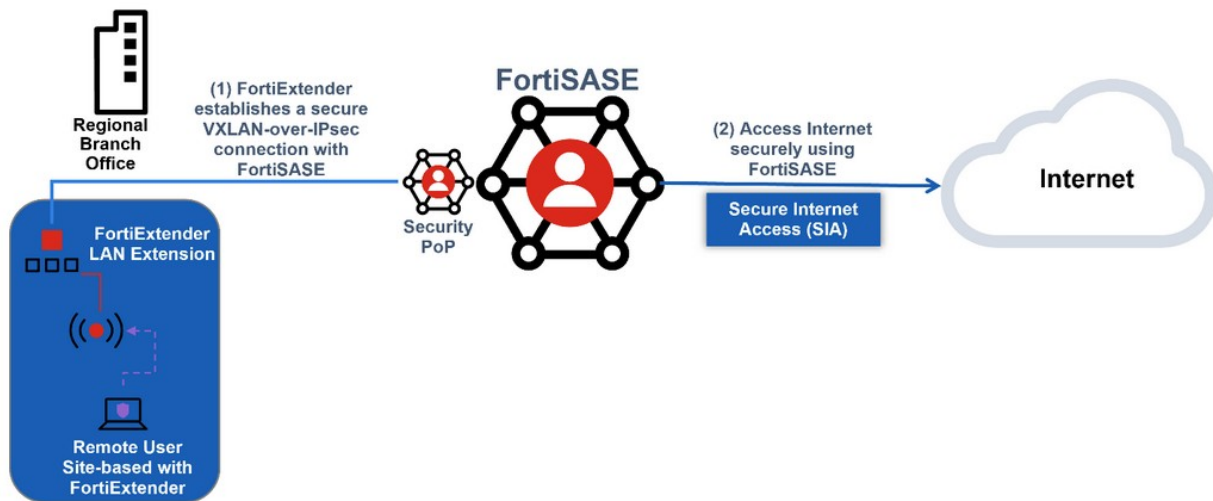
Figure B-4 — SIA for agentless remote users, typical topology (Fortinet original)



B.5 SIA for site-based users — FortiExtender LAN extension

Site-based SIA can use a FortiExtender as a LAN extension via a VXLAN-over-IPsec tunnel to FortiSASE, creating a Layer-2 network between FortiSASE and the site. Endpoints simply default-route to the FortiExtender — no FortiClient or proxy settings needed. Included for completeness; HCT's campuses use the higher-capacity Branch On-Ramp / FortiGate patterns below.

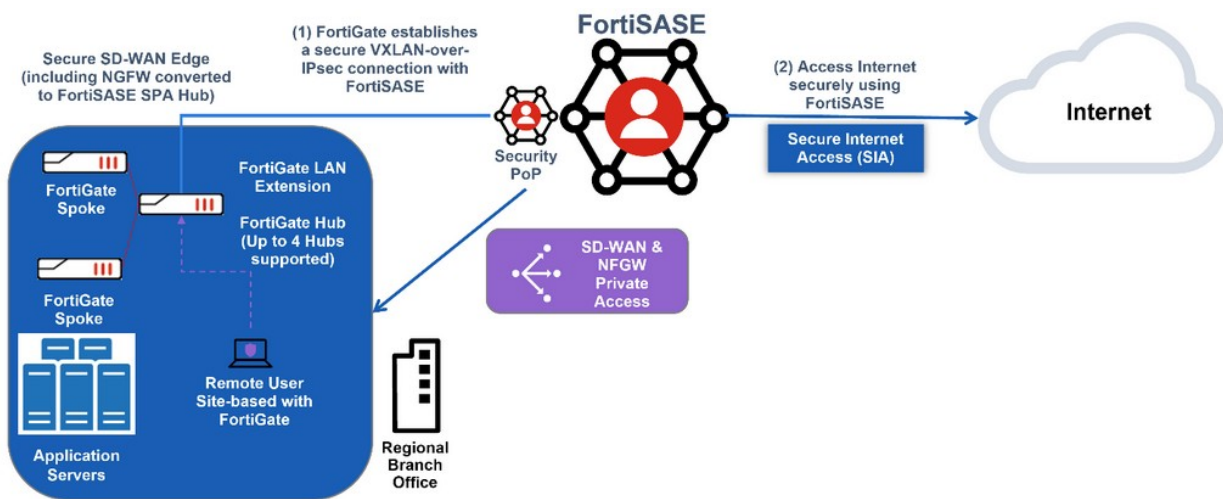
Figure B-5 — SIA via FortiExtender LAN extension, typical topology (Fortinet original)



B.6 Site-based users — FortiGate SD-WAN as secure edge → HCT campuses & DCs

A FortiGate SD-WAN device can act as the LAN extension / secure edge, centralising a site's connectivity to the FortiSASE FWaaS so endpoints only default-route to the FortiGate. The same FortiGate can later serve Secure Private Access as an SD-WAN or SPA hub. In the HCT design this pattern is delivered by the FortiGate-401F bundles at the Primary and DR data centres, with campus IPsec Branch On-Ramps carrying site traffic to the PoPs (Figure 4, Option 1).

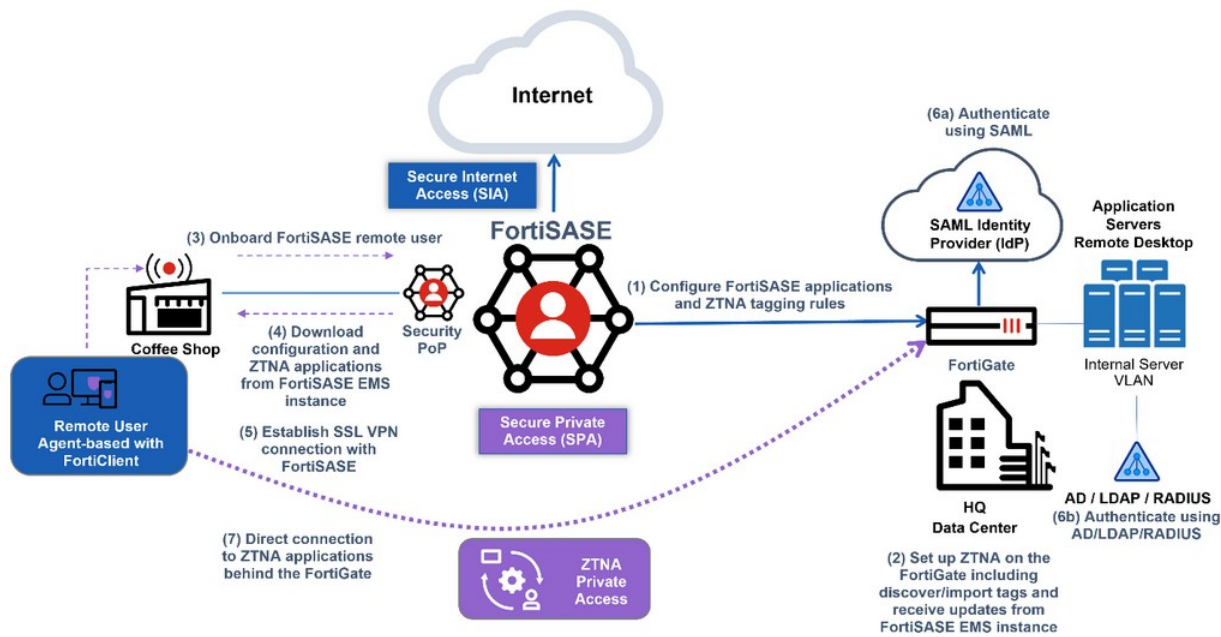
Figure B-6 — Site-based users via FortiGate secure edge, typical topology (Fortinet original)



B.7 Secure Private Access using ZTNA → HCT optional scope (Section 3.25)

Agent-based users securely access private TCP applications using ZTNA, taking the direct (shortest) path with per-session user authentication. A FortiGate at the organisation's data centre acts as ZTNA access proxy; FortiClient is managed by the FortiSASE Endpoint Management Service, which discovers device information, logged-on user and security posture, issues the client certificate, and applies ZTNA tagging rules. FortiSASE shares tags and certificate details with the FortiGate, which verifies identity and grants or denies access per tag. This is exactly the HCT optional-ZTNA design of Section 6 / Figure 5, with the FortiGate-401F pairs as access proxies.

Figure B-7 — Secure private access using ZTNA, typical topology (Fortinet original)



B.8 SPA using SD-WAN

Organisations with FortiGate SD-WAN deployments can give remote users broader access to privately hosted TCP- and UDP-based applications: the Security PoPs act as spokes in the SD-WAN network, using IPsec overlays and BGP to route between PoPs and the networks behind the SD-WAN hubs. FortiSASE PoPs and FortiGate hubs form a hub-and-spoke topology supporting Fortinet auto-discovery VPN (ADVPN), which lets spokes establish dynamic, on-demand shortcut tunnels; traffic is initiated from FortiSASE spokes toward FortiGate spokes.

Figure B-8 — SPA using SD-WAN, typical topology (Fortinet original)

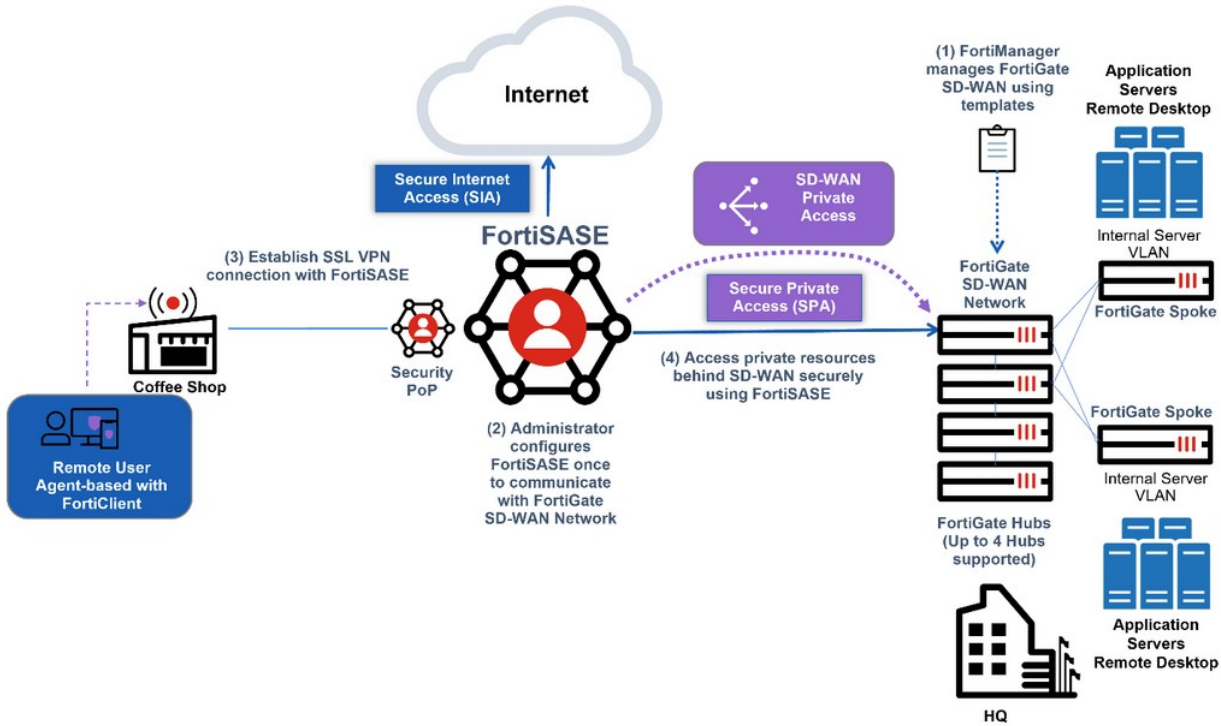
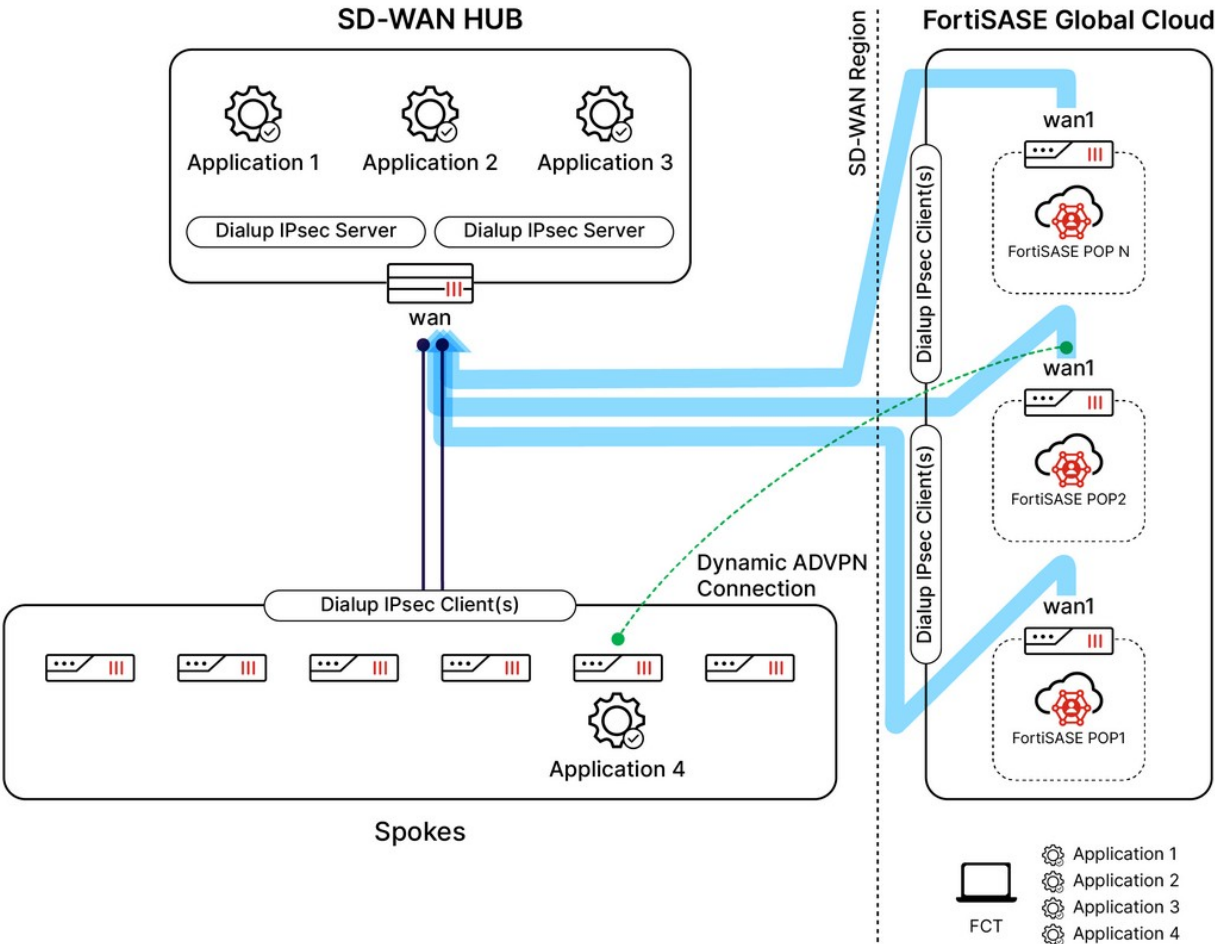


Figure B-9 — SPA using SD-WAN with ADVPN shortcut tunnels (Fortinet original)



B.9 SPA using NGFW

Organisations with FortiGate NGFW deployments convert the NGFW to a standalone IPsec VPN hub; the Security PoPs act as spokes, relying on IPsec overlays and BGP to route between PoPs and the networks behind the NGFW (up to four hubs). The same ADVPN hub-and-spoke behaviour applies. In the HCT design, the FortiGate-401F SPA Connectors included in the Option 1 and Option 2 bundles enable this pattern at the Primary and DR data centres.

Figure B-10 — SPA using NGFW, typical topology (Fortinet original)

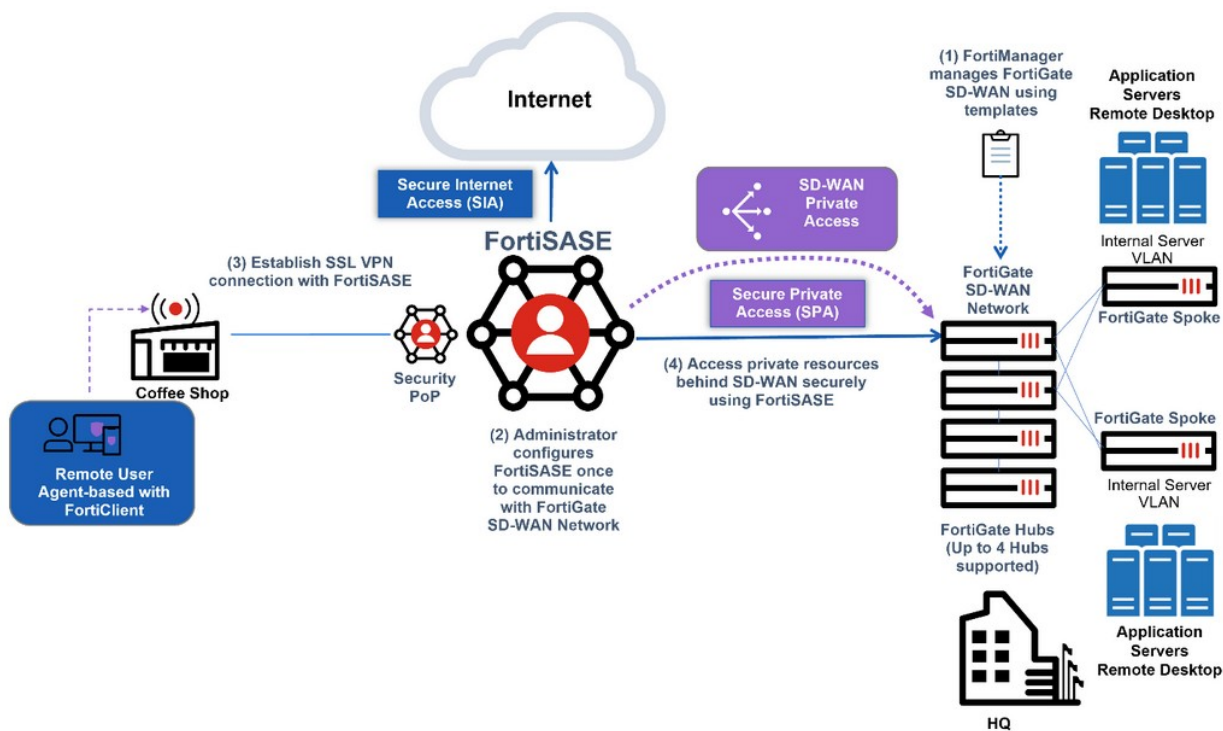
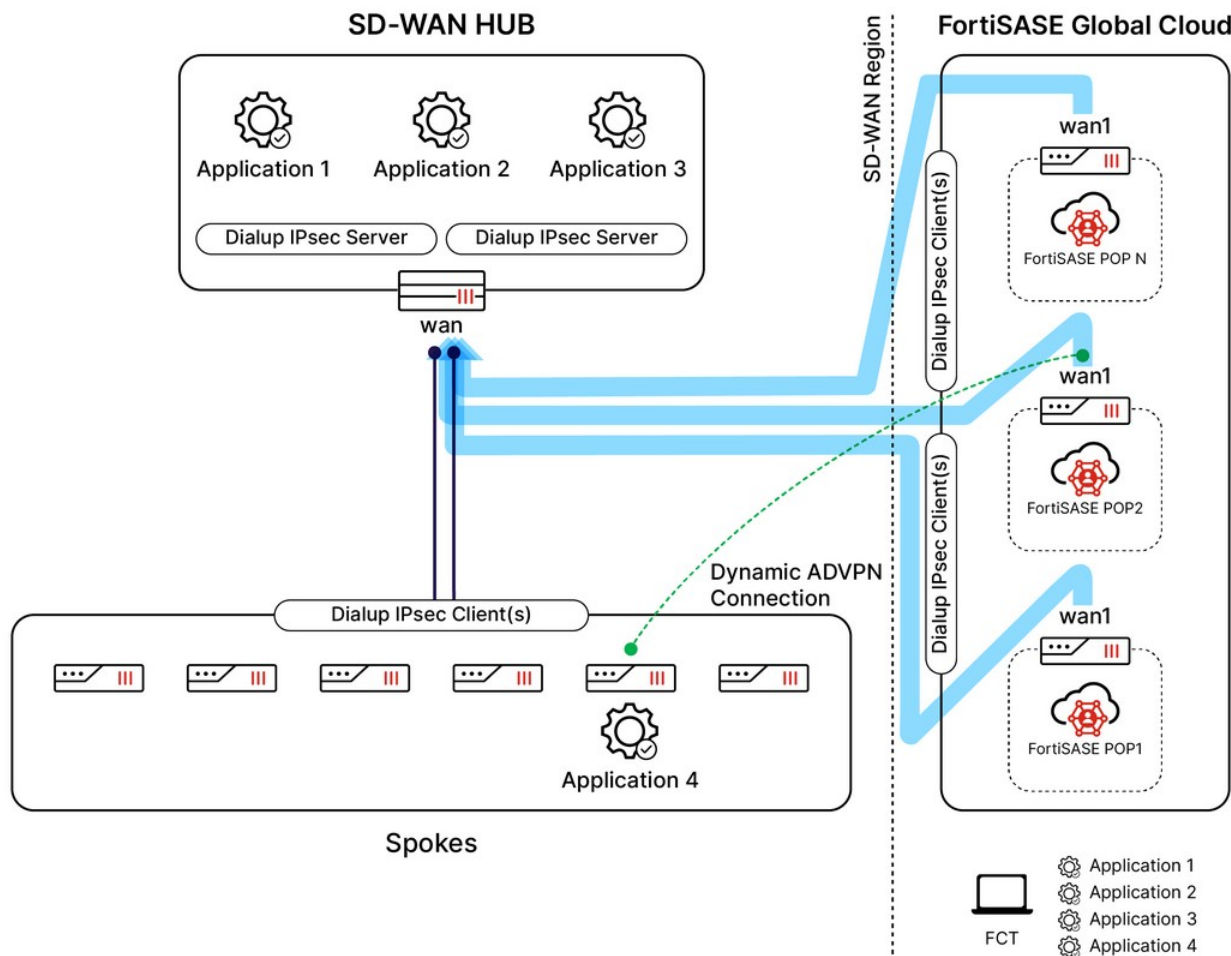


Figure B-11 — SPA using NGFW with ADVPN shortcut tunnels (Fortinet original)

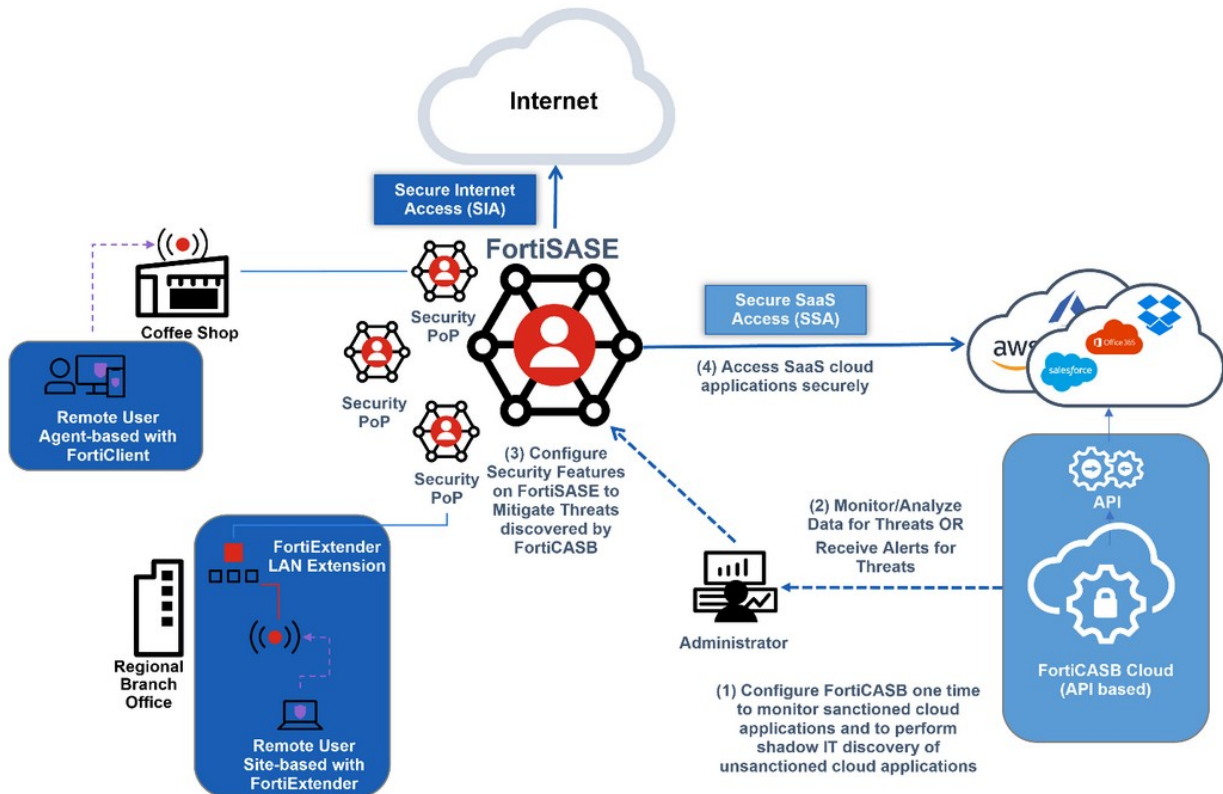


B.10 Secure SaaS Access — FortiCASB (API-based)

FortiCASB takes an API-based approach, pulling data directly from SaaS applications such as Office 365 using REST API queries with OAuth 2.0, enabling deep inspection of cloud data with advanced

monitoring, analysis, reporting and alerts on suspicious activity — out-of-band, with no impact on user SaaS performance. Access to FortiCASB is included with per-user FortiSASE licensing.

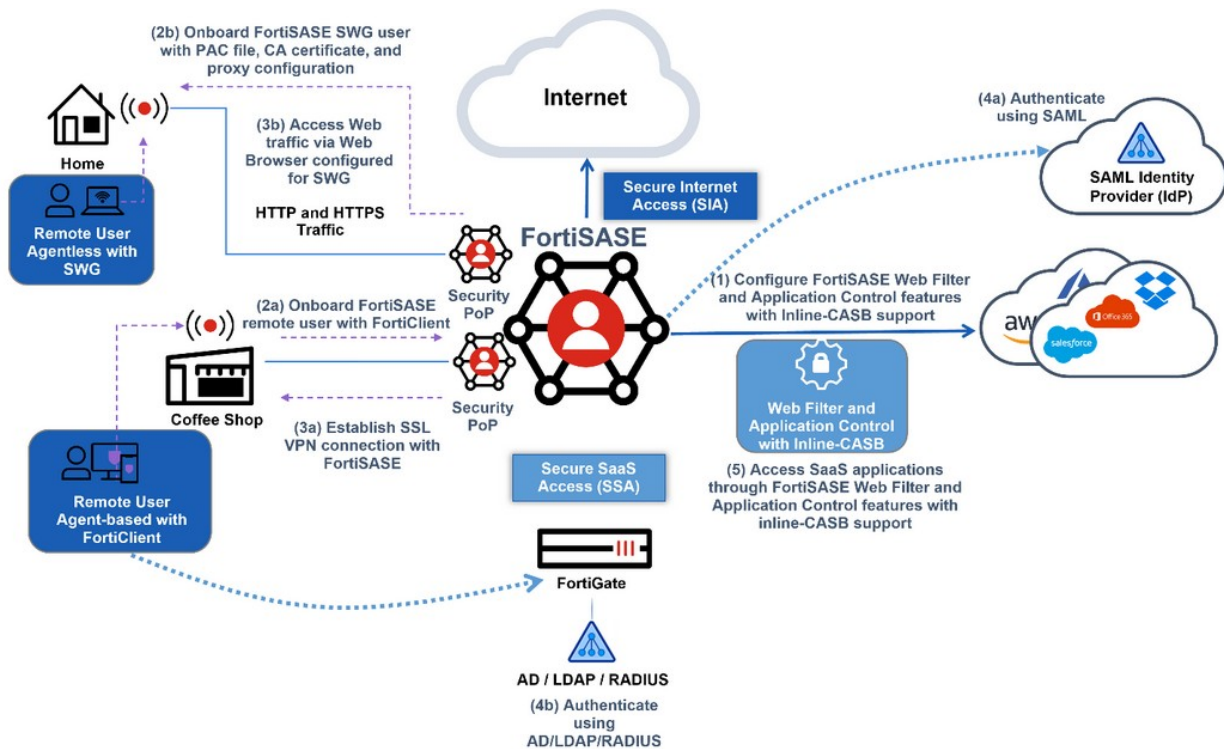
Figure B-12 — Secure SaaS access using FortiCASB, typical topology (Fortinet original)



B.11 Secure SaaS Access — FortiSASE Inline-CASB

FortiSASE provides Inline-CASB through its Application Control and Web Filter components with SSL deep inspection, sitting between users and their cloud services to enforce policy on SaaS traffic for both agent-based and agentless users. HTTP header insertion allows tenant-restriction control for SaaS applications. Inline-CASB requires no licence beyond per-user FortiSASE licensing — it is included in the proposed Advanced tier and drives the HCT DLP/tenant-control requirements of Sections 3.5 and 5.4.15.

Figure B-13 — SSA using FortiSASE Inline-CASB, typical topology (Fortinet original)



7. Why This Architecture Wins for HCT

- One platform, one agent, one console: eliminates the dual-platform operating model HCT runs today, cutting operational complexity while achieving BYOD/managed policy parity (Objective 1.6).
- Education-grade scale and rhythm: capacity surge, change freezes and readiness reviews are engineered around the academic calendar, not retrofitted to it.
- Compliance by design: platform ISO 27001 / SOC 2 Type II / CSA STAR; alignment with UAE IA (TDRA), NESAC, Federal Decree-Law 45/2021, NIST CSF 2.0, ISO 27001:2022, FERPA and PCI-DSS as applicable.
- Delivered by the UAE's Fortinet Expert partner for SASE and ZTNA, with OEM engineering directly behind the escalation matrix.

Architecture diagrams (HLD visual, per-flow diagrams and PoP map) are provided in the accompanying deck and will be formalised in the HLD/LLD deliverables in editable Visio/Draw.io formats per Section 3.13.7.